

SNAKE LU · TAIPEI

I design the architecture, and run an AI agent pipeline to ship it.

16 years in systems development and architecture, specializing in payment security, high-concurrency platforms and multi-tenant systems. I now work as an agentic engineer: I own the architecture, task breakdown and quality gates, while parallel AI agents handle implementation — one person delivering what used to take a team.

```
~/projects · zsh

$ git log --author=snake --since=2025-
10 --no-merges | wc -l
3,300+
$ claude --worktrees --parallel
✓ 44 worktrees · cross-model review on
$ report --month=2026-08
points 743  team share 60%  2.9× the #2
$ ls ~/Applications/home-cooked
MyGhost  MyXnip  TypeTop  AirGrip  +8
$ _
```

Systems Architect · Agentic Engineer · Toolsmith

ABOUT

Understand the business logic first — that is what makes a system solid

I've been building websites since 2010. Early roles at Yahoo! Super Mall, Wretch and PChome mobile payments (lead developer on the first version of Pi Wallet) gave me a foundation in e-commerce and payments. I then led overseas gaming platform teams in the Philippines and Taiwan — up to 21 engineers at once — running online platforms with weekly revenue above NT$100 million.

With a background in accounting, what I care about most at work is that the money is never wrong. Cross-system payment confirmation, timeouts and duplicate credits, reconciliation and recovery — I design all of these in from day one; handling the failure cases is often more than three times harder than the happy path. My recent work on two-phase transfer safety, withdrawal dispatch guards and load-test tuning all comes from this principle.

Since 2025 I've switched entirely to agentic engineering: Claude Code, Codex and Gemini form a reviewable, re-runnable development pipeline, while I focus on architecture, task breakdown, review and release gating. A small team can deliver what a large one does.

2.9×	62.8→0.03%	90s→2s	-96%	75 in 2 days	7 days
One month's output vs. the team's #2	Failure rate in a production-scale load test	Query on a 13.47M-row table	SQL statements for a 30-platform report	Third-party platform configs auto-extracted	AI product from kickoff to public beta

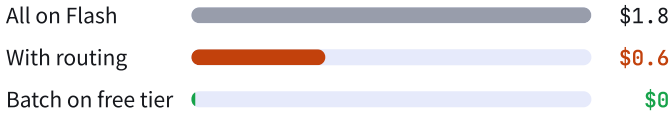
COST

Fluent in AI, sharper on cost: designing costs down from the start

US$0.6 / heavy user / month

LLM inference cost of an AI companion product

Estimate: 50 messages per user per day, 3K input / 300 output tokens; gross margin above 85% on a ¥30 monthly pass.



AI cost LLM

- **Route models by task**: Flash for live chat, Flash-Lite for all background work (2s measured), offline batches on a local CLI or free tier
- **Keep tokens lean**: send only the latest 40 messages verbatim and fold older ones into a rolling 80–200-character summary; output caps for thinking models tuned by testing so replies never truncate and waste a retry
- **No wasted quota**: rotate multiple keys and pause background jobs for 30 minutes when every key hits 429; built my own usage ledger and quota dashboard
- Moved the divination feature to Gemini's free tier: **$0 API cost**, latency about 45s → 20s

Development cost DEV

- **60% of the team's output from one person**: the agentic pipeline lets a small team ship at large-team volume
- **Tools instead of manual work**: configs for 75 third-party platforms auto-extracted in 2 days; API logging hooked at the HTTP layer so new vendors are covered with zero code changes
- The industry ranking tracker runs its Claude and ChatGPT tracks on subscription CLIs, with **zero API fees**

Traffic & cloud cost INFRA

- **Free traffic +58%**: after an SEO/AEO overhaul, 4-week Google impressions went from 12,093 to 19,163 with no paid ads
- Analytics at the edge: Cloudflare Workers + Analytics Engine collect events, batch-aggregated into MySQL without touching the primary database
- Graviton instances, with migration-credit (MAP) tags baked into IaC default tags so an apply never wipes the credits; voice via free Edge TTS, with synthesized lines cached permanently

US$1.80 / $0.60 are estimates from official list prices, not actual bills; other figures come from project code, configs and performance reports.

AGENTIC ENGINEERING

Not just using AI — running agents as a production line

In 2026 the industry calls this way of working agentic engineering: humans own architecture and quality, while AI agents write, test and ship. This is the pipeline I actually run every day.

01

Architect & break down

Pin down business rules and failure cases first, split requirements into independently verifiable units, and write the acceptance criteria.

`Owned by me`

02

Parallel implementation

One git worktree per ticket, so many agents develop at once without stepping on each other — up to 44 parallel workspaces on a single project.

`Run by AI agents`

03

Cross-review & diff testing

Claude writes, Codex reviews. Rewrites treat the legacy system's real output as ground truth, verified column by column with a custom black-box diff tool.

`Cross-model + tooling`

04

Gate & release

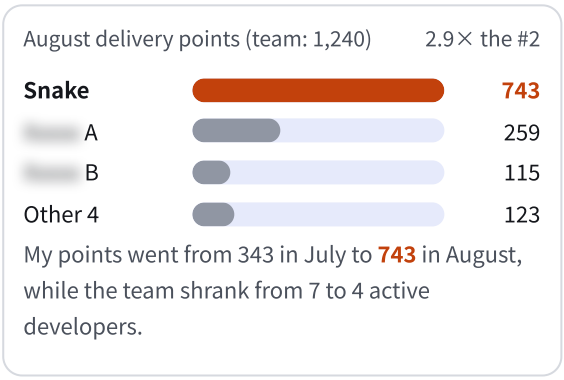
Human review, tests and canary flags before anything reaches production; the process is captured as re-runnable skills and docs.

`Owned by me`

Proof: the month the team shrank from 7 to 4, the project still launched on time

In August 2026, three teammates left one after another, leaving 4 active developers. I took on cross-track development, handovers and three-environment releases, and we launched on schedule on September 1.

- Delivered 60% of the team's story points and 56% of merged MRs single-handedly, across five tracks: backend, admin, frontend, store and cloud
- Integrated 6 game providers in one week, each through a three-stage rollout with live testing
- Fixed 12 production ops errors in the week before launch and ran the three-environment release solo
- Wrote 31 assessments, specs and handover documents the same month — about 220,000 Chinese characters

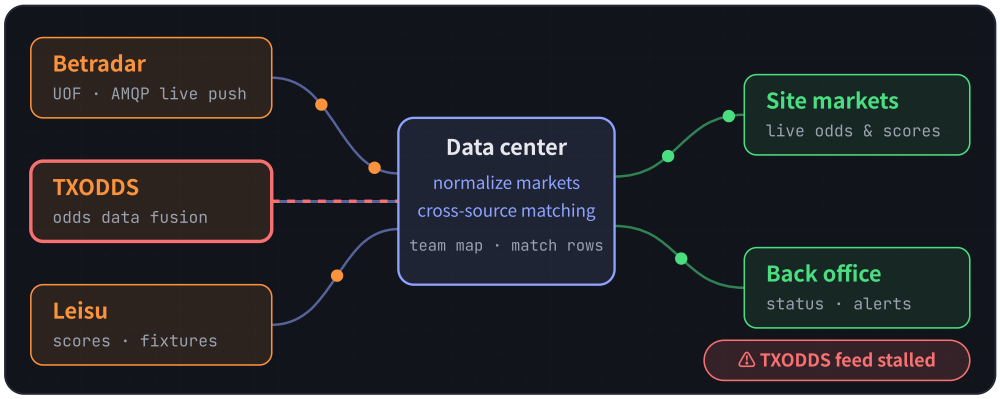


DATA SOURCES

A year of sports data feeds: three sources, one data chain

Owned ingestion, normalization, cross-source matching and monitoring for three sports data providers — Betradar, TXODDS and Leisu — carrying data from source all the way to each site's markets and the main back office.

3	5,072	9	1 yr
Betradar · TXODDS · Leisu	odds updates per hour (measured)	markets per match, live at open	of data feed development



Live ingestion & normalization

Betradar UOF subscribes to the full odds push over AMQP with recovery re-fetch; TXODDS odds fusion plus Leisu scores and fixture master data. All three formats are unified into the data center's events, markets and outcomes. **Measured 5,072 odds updates in one hour, with 9 markets per match landing live at open.**

Cross-source event matching

With Leisu fixtures as the master, a team mapping table ties every source to the same match and produces multi-source rows for sites to read. **Found why Betradar failed to match: knockout fixtures were created with bracket placeholder teams, and the mapping was never updated after the teams advanced and were renamed.** Matching recovered on its own once the mapping was added.

Monitoring, alerts & troubleshooting

Built stalled-feed alerts: when a source stops updating but the market is still open, it is flagged "market stalled", linking Betradar/TXODDS → data center → main back office. Checked missing-market complaints against the Betradar CTRL monitoring console, confirmed lower-tier cup matches only open 4–12 hours before kickoff, and turned it into a diagnostic checklist and support reply template.

Figures come from the production database and incident investigation records; client and site names are omitted for confidentiality.

RECENT WORK · 2025–2026

Representative projects from the past year

Client projects are described by product type for confidentiality.

FULL-TIME

Game API aggregation platform: risk control & new-market wallet

2020.04 – present

Lead developer for risk control and the merchant back office. Built an SMS-invite marketing system from scratch (Cloudflare Workers + Analytics Engine), a time-zone architecture spanning 8 time zones × 4 systems, and the internationalization of a trilingual South Asian wallet.

Large-table query 90s → 2–3s · new system live in one month

PHP · ThinkPHP · Cloudflare Workers · Redis · Nuxt

CONTRACT

White-label entertainment platform: wallet, payments & cloud

2026.07 – present

Two-phase transfers with timeout reconciliation and withdrawal dispatch guards; WAF allowlists pushed directly by the app and written back to Terraform; a production-scale load test surfaced three root causes. 788 commits in three months — 29% of the team's total.

Failure rate 62.8% → 0.03% · p95 60s → 10.9s

Go · AWS EKS · WAF · Terraform · Helm · Vue 3

Multi-tenant platform: affiliate commission back-office API rewrite

2025.10 – 2026.09

Rewrote legacy PHP into a three-layer Lumen 10 architecture, writing 73 of the 86 routes; report aggregation across 70+ schemas; config-driven support for 75 platforms at once; a declarative diff-sync engine that passed client acceptance.

SQL statements -96% · 41 of 42 platforms match old vs. new column by column

PHP 8.3 · Lumen · PHPUnit · MySQL · Redis

Sports data integration: Betradar • TXODDS • Leisu

2025.10 – 2026.09

One year of sports data feed development. Integrated Betradar UOF (live odds push over AMQP), TXODDS odds and Leisu scores and fixture master data, imported into a data center and matched across sources for each site. Built stalled-feed alerts: when data stops updating while a market is still open, it is flagged "market stalled", linking Betradar/TXODDS → data center → main back office. Traced and fixed broken cross-source matching caused by bracket placeholder teams in knockout rounds; investigated missing-market complaints against the Betradar CTRL monitoring console.

3 data sources · stalled-feed alerts · cross-source matching

PHP · Laravel · AMQP · Betradar UOF · MySQL · Job monitoring

Contractor payment slips & general ledger system

2026.03 – present

Moved contractor payment slips online for three companies in a group: a tax-rule engine (including the 183-day residency rule), four-level approvals, 20,000-row batch uploads and bilingual PDFs. The follow-on accounting system has finished requirements interviews and a four-phase plan.

3 APIs + 3 frontends · delivered and still growing

PHP 8 · Next.js 16 · TypeScript · MariaDB

Full-stack platform for an astrology brand

2026.04 – present

Content site, membership and interactive divination tools, built solo. AI drafts, experts review before publishing, and the generation worker is sandboxed. After an SEO/AEO overhaul the core keyword ranks #1 on Bing and Google impressions rose 58%.

121 APIs · zero-downtime domain migration

Next.js · PHP · Three.js · Gemini ⇄ Claude · LINE Login

OWN PRODUCTS

NianNian: AI character companion platform

2026.09

Layered memory, proactive messages driven by real presence heartbeats, a 6-stage relationship arc and multi-character group chat. Conversations stream through Gemini while other generation runs on a local Claude to keep costs down.

7 days from kickoff to public beta

Node.js · SQLite · SSE · Gemini · Claude CLI

Rankly: industry website rankings & AI visibility

2026.08 – present

Daily health checks on clients' and competitors' sites, plus real questions put to ChatGPT, Claude, Gemini and Perplexity to rank who AI actually sees; includes a five-point GEO diagnosis and daily content trends.

857 AI answers in 45 days · found the bottleneck is coverage, not rank

Node.js · Multi-LLM · headless Claude · GEO

TOOLSMITH · HOME-COOKED SOFTWARE

I build my own tools

The ideal team in *The Mythical Man-Month* includes a "toolsmith" who builds the tools everyone else relies on. AI agents make that cheap: rather than living with the limits of commercial software, build one that fits your own workflow. In 2026 I built 12 native macOS apps, and I use every one of them daily. Public ones link to their GitHub downloads.

Customized Ghostty fork

MyGhost

A terminal built for long-running AI agents: sidebar tabs, tmux so sessions survive crashes, auto-reconnecting remote SSH tabs, pasted images auto-uploaded to remote agents, a Claude/Codex quota dashboard, plus a mobile-friendly web version.

Download latest on GitHub →

Own build, Xnip-style UI

MyXnip

The old screenshot API could freeze the whole machine, so I redesigned it on ScreenCaptureKit as capture first, display later. Scrolling captures, annotation, OCR.

Rewrite of an existing app

TypeTop

Hold right ⌘ and talk: Whisper transcribes, an LLM cleans it up and types it into any text field. Supports on-device Apple Intelligence and automatic failover across 7 LLM providers.

Download latest on GitHub →

Own build, modeled on Going IME

SuperNatural IME

A custom Chinese input engine on InputMethodKit: DP word-lattice composition, selection learning with decay, abbreviation shortcuts.

Original

AirGrip

Control your Mac with hand gestures through the camera, using Vision hand-pose detection — all on-device.

Customized DuoBar fork

MyDuoBar

Menu-bar status hub: Wi-Fi, VPN, per-earbud AirPods battery and the native charge limit, under Swift 6 strict concurrency.

Download latest on GitHub →

Own build, Downie-style UI

My Downie

Video downloader that covers platforms the original can't; one-click launch from a browser extension, and the app updates itself.

Own build, Niya-style UI

My Niya

A Dynamic Island-style dashboard at the top of the screen: system monitor, to-dos, network speed, clipboard and AirDrop.

Own build, inspired by Hidden Bar

MyHiddenBar

Worked out the new menu-bar layout rules in macOS 27 and rewrote the hiding logic — done in a day.

Own build, Obsidian-style UI

MyObsidian

Markdown knowledge base: backlinks, mermaid, cross-platform builds and e2e tests.

Download latest on GitHub →

Own build, Slidepad-style UI

MySlidePad

A browser panel that slides out when you touch the screen edge, with scripts for organizing AI chats.

Download latest on GitHub →

Original

MyPlayer

A lightweight player with keyboard control of speed and looping.

SKILLS

Skills

AI & Agentic

Claude Code / Codex / Gemini

Parallel multi-agent dev on worktrees

Cross-model review • reusable skills

LLM product integration & cost routing

GEO / AEO

Backend & Data

PHP (Laravel • Lumen • ThinkPHP)

Go (Gin • GORM)

Node.js

MySQL / MariaDB tuning • Redis

Queues • scheduling • idempotency

Frontend & Client

Next.js • Nuxt • Vue 3

TypeScript • Tailwind

Swift • SwiftUI • AppKit

PWA • Three.js

i18n

Cloud & Architecture

AWS EKS • WAF • CloudFront • RDS

Terraform • Helm • GitLab CI

Cloudflare Workers • Tunnel

High concurrency • load testing • canary

Payment security & reconciliation

EXPERIENCE

Experience

2020.04 – present

Overseas game API aggregation platform

Engineering Team Leader • Taipei

Extended Laravel and ThinkPHP systems and led R&D in building the collection-merchant architecture and operations back office; integrated third-party payments including Alipay, bank cards, mobile top-up and USDT; led cross-border communication (Philippines, Dubai) and remote collaboration. Recently responsible for risk control, the merchant back office, marketing systems and new-market wallets.

2025 – present

Contract development

Systems architecture • Full-stack • Agentic engineering

Enterprise platform rewrites and embedded development (affiliate commission back office, wallet payments and cloud), plus from-scratch delivery for SMBs and brands (contractor payment & accounting system, astrology brand platform).

2019.05 – 2020.01

Taiwan gaming fintech company

Engineering Team Leader • Taipei

Built third-party game platform integrations in Yii and Go, owning game logic and financial reporting; developed a real-time routing and hedging system for large orders in Go.

2017.07 – 2019.01

Desheng Group (Philippines)

Deputy Director of Engineering • Taipei

Sent from the Manila headquarters to build a team in Taiwan, leading up to 21 people. Built a user analytics center, game center, top-up, risk control and revenue-sharing system on Laravel; led the team at G2E Asia 2018 in Macau.

2016.04 – 2017.06

GamePark Group (Philippines)

Team Leader • Manila

Ran online platforms averaging over NT$100 million in weekly revenue: built one new platform and operated four; shared multi-platform core, NoSQL load testing and mobile game APIs.

2014.11 – 2016.04

PChome Online

Team Leader • Taipei

Lead developer on the first version of the Pi mobile wallet: multi-host architecture, a database caching package, bank payments, risk control and accounting; introduced testing and Scrum.

2010 – 2014

Yahoo! • Tripmoment • Interfly

Web Development Engineer • Taipei

Purchase data analysis and ad partnerships for Yahoo! Super Mall; refactored Wretch (test coverage 70%→80%); app APIs for a travel startup; a government website that won Taiwan's Golden Finger Award.

CONTACT

Need a system made solid, or want AI in your workflow?

Full-time roles, consulting and contract work are all welcome — just email me. Online: resume-snake.master0987.com/en

snakeluph@gmail.com

GitHub